

WP-R 369 Signature Channel — User & Technical Manual

Version 2.0.1. Covers installation, routing, the full control reference, metering, and troubleshooting.

1. What it is

WP-R 369 Signature Channel is DMI Track's in-house ribbon-microphone and output-transformer model. It is not an EQ/compressor channel strip — there is no EQ band anywhere in the signal path and no gain-reduction dynamics processor. What it actually contains is a stateful "ribbon element" model (a velocity/momentum/damping integrator feeding a tanh-style saturator, run independently per channel with memory that persists sample-to-sample) followed by a separate output-transformer/iron-core stage with its own drive, magnetic-flux, hysteresis-memory and lamination modeling, a final multiplicative gain stage, and a fixed safety ceiling. Every internal curve and default value is derived from the golden ratio ($\phi \approx 1.618$) rather than arbitrary decimals — the shared `DIH_ENGINE_CLEAN` "canon" headers this plugin builds against enforce that constraint at compile time via `#error` checks in `DIHGuard.h`.

Signal flow, in order, per block: **(1)** channel-mode handling — Mono sums L+R before anything else runs, Dual Mono + Independent skips the shared stereo stages entirely; **(2)** a proximity-effect gain boost applied to the input ahead of the ribbon stage (skipped in Dual Mono + Independent); **(3)** the pressure-responsive mechanical ribbon stage, whose signal-following pressure envelope drives Damping, Excursion, and the selectable Ribbon Guard; **(4)** the ribbon element itself, run per channel with persistent velocity/momentum/mass/contour state; **(5)** a mid/side polar-pattern stage (Width/Focus) plus a second golden-ratio-scaled mid/side voicing keyed to the Polar Pattern choice (Figure-8/Cardioid/Omni), which also re-biases the ribbon stage's own Width/Focus/Convergence/Clarity ranges before step 4 runs; **(6)** the transformer/iron-core stage — flux-driven core saturation, an optional 3rd/6th/9th-harmonic cascade ("Imhotep369", the source of the "369" in the product name) when Lamination is set to Phi Toroid, a one-sample hysteresis memory feedback loop, and a secondary load-dependent nonlinear drive pass; **(7)** a final gain stage combining Output Trim, Core Headroom, Sec Coloring and Iron Geometry, then the Output fader in dB, with the same fixed safety ceiling ($\phi/2 \approx 0.809$ linear, $\approx \blacksquare 1.8$ dBFS) reapplied after every stage. That ceiling is a hard limiter for overload safety only — source comments explicitly note it must not behave as an always-on musical compressor, and there is no attack/release or gain-reduction metering anywhere in the plugin.

2. Installing

The signed release lives under

`Releases/DMI_TRACK_WP-R_369_Signature_Channel_RELEASE_<date>/Individual/:`

- `DMI_TRACK_WP-R_369_Signature_Channel_2.0.1_Installer_<date>.dmg` — disk image
- `DMI_TRACK_WP-R_369_Signature_Channel_2.0.1_Installer_<date>.pkg` — signed installer package (inside the DMG), installs:
- `WP-R 369 Signature Channel.component` → `~/Library/Audio/Plug-Ins/Components/`
- `WP-R 369 Signature Channel.vst3` → `~/Library/Audio/Plug-Ins/VST3/`

There is no standalone build — `CMakeLists.txt` declares `FORMATS AU VST3` only, no `Standalone` target. If you'd rather place the bundles yourself instead of running the `.pkg`, copy the two items above to the same two folders by hand.

Quit your DAW before copying a new `.component` over an old one — overwriting the binary of a plugin instance that's currently loaded can crash the host or leave it running a stale copy.

If a newly installed version doesn't show up, or the host still behaves like the old build after copying the new one, the `AudioComponent` registry cache is almost always the cause:

```
``bash killall -9 AudioComponentRegistrar ``
```

No logout or restart needed. If that alone doesn't do it, also clear the host's own AU cache (in Logic: `rm -rf ~/Library/Caches/AudioUnitCache`) and relaunch.

The plugin seeds its own factory preset folder (`~/Library/Audio/Presets/DMI_TRACK/WPR_MKVI/`) the first time it loads — no manual preset install step is needed. The repo also ships `install_wpr_presets.sh`, which just creates that folder

ahead of time and reports how many .dih files are in it; it's a convenience/diagnostic script, not a required part of installation.

3. Loading / routing

`isBusesLayoutSupported()` accepts two shapes:

- **Matched in/out layouts**, up to 8 channels each (mono/mono, stereo/stereo, or wider layouts such as quad/quad, all accepted structurally). The DSP itself only ever reads and writes channels 0/1 (L/R) — a source comment confirms this directly: any channels beyond the first two in a matched wide layout pass through completely untouched.
- **Mono in → stereo out**. The mono input is copied to both channels before processing, so the plugin behaves like a mono source being picked up and rendered through the stereo ribbon/polar/transformer chain.

Any other mismatched layout (e.g. stereo in → mono out) is rejected.

Within a stereo instance, the **Channel Mode** parameter adds a second, independent layer of routing control on top of the host bus layout: MONO force-sums L+R before any processing regardless of the bus layout; STEREO (default) is the normal two-channel path with the shared proximity/polar stage; DUAL only changes behaviour when **Dual Mono Link** is also set to INDEPENDENT, in which case the two channels run through completely separate ribbon states with no proximity sum and no shared mid/side polar stage — true dual-mono processing.

4. Interface

The editor is resizable (`setResizable(true, true)`), with limits of 900×579 up to 1400×900, opening by default at 960×617. Layout:

- **Top chrome bar** — category tag, the preset dropdown with Prev (<) / Next (>) / Reset buttons, and a status readout on the bottom bar showing the current preset name plus a static WP-R 369 MKVI | AU / VST3 tag.
- **Choice plate** (two rows of four dropdowns) — Lamination, Phi Family, Core, Winding on the top row; Age, Polar Pattern, Channel Mode, Dual Mono Link on the second.
- **KNOB LEVEL quick-set row** — three buttons labeled 9, NOON, 3 (clock-face positions). Each one jams all nine ribbon-element knobs (Transformer, Ribbon Tone, Mass, Side Width, Polar Focus, Output Trim, Air Edge, Ribbon Blend, Proximity) to 25%, 50%, or 75% of the plugin's internal safety ceiling in a single click. It does not touch the transformer/iron-core section, Polar Pattern, Channel Mode, or Output.
- **3×3 rotary knob grid** for the nine ribbon-element controls, with the Output fader centred below it.
- **Advanced physics strip** for Ribbon Damping, Ribbon Excursion, and Ribbon Guard. These controls are signal-dependent mechanical response controls, not a hidden EQ or compressor.
- **Stereo Field Visualizer** — a phi-proportioned circular plot on the right side of the canvas (see §6).
- **Peak level bar** along the right edge of the window (see §6).
- Background artwork is drawn from an embedded texture (`WPR.png`), layered at two different opacities behind the controls.

Controls

All ranges and defaults below are taken directly from `createLayout()` in `source/PluginProcessor.cpp`. The `Control` column gives the on-screen/automation name; the parameter ID used by the host and by scripting is in backticks.

Ribbon Element

Control	Range	Default	Notes
Transformer (DRIVE)	0.0 – 1.0	0.404508	Drive into the ribbon element's own tanh saturator. Despite the name, this is the ribbon's saturation, not the output transformer — see Iron Drive below.
Ribbon Tone (ARCHITECTURE)	0.0 – 1.0	0.471927	Sets the damping of the ribbon's velocity/momentum integrator — higher values settle faster with less resonant tail.

Control	Range	Default	Notes
Ribbon Mass (DEPTH)	0.0 – 1.0	0.404508	Simulated inertial mass of the ribbon. Higher mass gives a slower, heavier low-mid response.
Air Edge (CLARITY)	0.0 – 1.0	0.471927	Weight of a high-frequency edge/transient term derived from the gap between the input and the ribbon's settled momentum.
Ribbon Blend (MIX)	0.0 – 1.0	0.471927	Dry/wet blend of the ribbon element only. At 0 the ribbon nonlinearity is bypassed; the transformer and output stages still run on the dry signal.
Proximity (CONVERGENCE)	0.0 – 1.0	0.269672	Drives a proximity-effect level boost applied before the ribbon stage, and re-biases the Width/Focus/Clarity ranges whenever a Polar Pattern other than the default is engaged. Has no effect in Dual Mono + Independent mode.

Mechanical Pressure / Excursion

The following controls sit before the stateful ribbon element and respond to the incoming pressure envelope. They are deliberately price- and model-agnostic: WPR cannot identify a microphone from audio alone and does not claim a hidden flat-response correction.

Control	Range	Default	Notes
Ribbon Damping (DAMPING)	0.0 – 1.0	0.618034	Controls recovery of the mechanical pressure state. Higher values settle the response faster; lower values retain more movement.
Ribbon Excursion (EXCURSION)	0.0 – 1.0	0.236068	Scales the pressure-driven ribbon travel and transient movement before the ribbon-element stage.
Ribbon Guard (RIBBON_GUARD)	Off / Soft / Medium / Hard	Off	Limits the rise-driven excursion response progressively. It is a protective mechanical guard, not a musical compressor with attack/release controls.

Stereo Field & Polar Pattern

Control	Range	Default	Notes
Side Width (WIDTH)	0.0 – 1.0	0.471927	Side-channel gain in the mid/side polar-reconstruction stage that runs after the ribbon element.
Polar Focus (FOCUS)	0.0 – 1.0	0.471927	Side-channel attenuation that narrows the picked-up field toward mono as it increases.
Polar Pattern (POLAR_MODE)	Figure-8 / Cardioid / Omni	Figure-8	Re-biases Width/Focus/Convergence/Clarity toward a pattern-appropriate range before the ribbon stage runs, then applies an additional golden-ratio-scaled mid/side gain pass after it. No effect in Dual Mono + Independent mode.
Channel Mode (channelmode)	Mono / Stereo / Dual	Stereo	Mono force-sums L+R before any processing. Dual only changes behaviour together with Dual Mono Link = Independent.

Control	Range	Default	Notes
Dual Mono Link (dualmonoLink)	Linked / Independent	Linked	Independent (with Channel Mode = Dual) runs two fully separate ribbon chains with no shared proximity or polar stage. Linked processes normally even when Channel Mode = Dual.

Transformer & Iron Core

Control	Range	Default	Notes
Iron Drive (IRON_DRIVE)	0.0 – 1.0	0.381966	Bipolar drive into the transformer core-saturation stage — separate from the ribbon element's own Transformer/Drive control.
Magnetic Flux (FLUX)	0.0 – 1.0	0.471927	Scales the core-saturation gain; weighted by Core Material.
Core Headroom (HEADROOM)	0.0 – 1.0	0.618034	Trims how strongly the iron stage colors the signal before the final gain/ceiling stage.
Primary Bias (PRI_BIAS)	0.0 – 1.0	0.500000	Biases a secondary, post-iron nonlinear drive pass (a second momentum-bend saturation stage).
Sec Coloring (SEC_COLOR)	0.0 – 1.0	0.236068	Secondary-winding coloring trim folded into the final multiplicative gain stage.
Winding Load (LOAD)	0.0 – 1.0	0.381966	Controls how much of the secondary nonlinear drive pass blends in; weighted by Winding Topology.
Permeability (PERMEABILITY)	0.0 – 1.0	0.618034	Scales how strongly the iron/hysteresis path blends against the clean ribbon signal.
Hysteresis (HYSTERESIS)	0.0 – 1.0	0.145898	Amount of core "memory" fed back from the previous sample's transformer state; weighted by Transformer Family.
Iron Geometry (GEOMETRY)	0.0 – 1.0	0.471927	Core geometry trim folded into the final multiplicative gain stage.
Phi Lamination (LAMINATION)	Phi Shell / Phi Toroid	Phi Shell	Phi Toroid adds a fixed-depth 3rd/6th/9th-harmonic cascade into the core-saturation path; Phi Shell leaves it off.
Phi Family (TRANSFORMER_FAMILY)	Phi A / Phi B / Phi C	Phi A	Weighting table applied to the hysteresis memory term.
Phi Core (CORE_MATERIAL)	Phi Nickel / Phi Steel / Phi Mu	Phi Nickel	Weighting table applied to the flux-drive gain.
Phi Winding (WINDING_TOPOLOGY)	Phi Bifilar / Phi Layer / Phi Inter	Phi Bifilar	Weighting table applied to the secondary nonlinear-drive blend amount.
Phi Age (CORE_AGE)	Phi New / Phi Aged / Phi Legacy	Phi New	Weighting table applied to the core-saturation output level.

Output

Control	Range	Default	Notes
Output Trim (ENERGY)	0.0 – 1.0	0.471927	Sets the clean-signal contribution to the final multiplicative gain stage — higher values raise the pre-ceiling level.

Control	Range	Default	Notes
Output (OUTPUT)	■ 12.0 dB – +12.0 dB (0.01 dB steps)	0.0 dB	Final trim, applied after a small fixed factory calibration constant (≈ ■ 0.1 dB) and re-limited by the same safety ceiling used throughout the chain.

30 parameters in total, all automatable and all recalled from session/DAW state via the plugin's `AudioProcessorValueTreeState`.

5. Presets

There are two separate, independently-addressable preset systems in this plugin:

Host program list. `getNumPrograms()` returns 81 — 27 base recipes × three channel-mode banks (Stereo, Mono, Dual Mono), listed by the host as e.g. `VOCAL INTIMACY [Stereo]`, `VOCAL INTIMACY [Mono]`, `VOCAL INTIMACY [Dual Mono]`. Selecting one from the host's own preset/program menu (Logic's plugin-header preset list, a VST3 program list, etc.) applies the full recipe — all ribbon/transformer parameters, Polar Pattern, and the matching Channel Mode/Dual Mono Link — in one call.

In-GUI preset shell. The plugin's own top-bar dropdown plus `</>` / `RESET` buttons is a real, working preset browser backed by individual `.dih` files on disk at `~/Library/Audio/Presets/DMI_TRACK/WPR_MKVI/`. On first run, if that folder is empty, the plugin seeds it with the same 27 factory recipes (Stereo-bank values only — the channel-mode expansion is specific to the host program list above). `</>` step through that on-disk list; `RESET` reloads whichever preset sorts first alphabetically in the folder — not necessarily "WP-R MKVI CANON" specifically. The processor also implements save and delete for this on-disk list (`saveCurrentPreset()` / `deleteCurrentPreset()`), but the currently visible interface doesn't expose them: a separate `PresetBar` component with its own `SAVE/DEL` buttons is constructed and wired to those same methods, but it is built hidden (`setVisible(false)`) and non-interactive (`setInterceptsMouseClicks(false, false)`), so it isn't reachable from the shipping UI.

6. Metering

Peak level bar — a single vertical bar along the right edge of the window, filled bottom-up. It's fed 30 times per second from the current block's peak output sample run through one further "momentum bend" nonlinearity; it's a relative peak indicator only, with no numeric dB readout, no RMS/LUFS mode, and no ballistics beyond that one shaping curve. The mechanical pressure controls do not replace this meter; use the bar and the host meter together when setting level.

Stereo Field Visualizer — the circular phi-proportioned plot to the right of the knob grid. It combines two things redrawn 30 times per second: a pattern outline reflecting the current Polar Focus/Side Width knob values (what the selected pickup pattern "looks like"), and a single dot plotted at the block's instantaneous mid/side peak position, showing where the output is sitting in the mid-side plane in real time. There is no gain-reduction meter anywhere in the plugin — there is no compressor to meter.

7. Building from source

Requires JUCE and the canonical DIH engine checked out locally. Unlike some sibling projects, this plugin's `CMakeLists.txt` resolves both paths from `$HOME` directly and does not read `-D` overrides for them:

- JUCE: `$HOME/Development/JUCE`
- DIH engine: `$HOME/Development/DIH/DIH_ENGINE_CLEAN` — configure fails with `FATAL_ERROR` if `Core/DIHWorl.d.h`, `Core/FieldState.h`, `Core/Imhotep369.h`, `Core/DIHGuard.h`, or `Engine/DIHEngine.h` are missing from it.

```
``bash cmake -S . -B build cmake --build build -j8 ``
```

Universal macOS binary (arm64 + x86_64), macOS 10.14 deployment target. There's no `ctest` suite defined in this project's `CMakeLists.txt`.

To run Apple's own validator once the AU is installed (`AU_MAIN_TYPE` is `kAudioUnitType_Effect`, i.e. type `aufx`):

```
``bash auval -v aufx WPR6 DMI1 ``
```

8. Troubleshooting

Symptom	Cause	Fix
Host still shows the old version, or a newly installed build doesn't appear	macOS AudioComponent registry cache	<code>killall -9 AudioComponentRegistrar</code> ; if that's not enough, also clear the host's AU cache (Logic: <code>rm -rf ~/Library/Caches/AudioUnitCache</code>) and relaunch.
macOS warns the installer is from an "unidentified developer" or blocks it outright on first open	Gatekeeper's on-disk assessment of a freshly downloaded, quarantined file can differ from the notarization status recorded at build time	Right-click the <code>.pkg</code> → Open, or clear the quarantine flag (<code>xattr -dr com.apple.quarantine <path-to-pkg></code>) and reinstall.
Harsh/distorted transients at extreme settings, especially on vocals	Stacking Iron Drive, Magnetic Flux, and Hysteresis all high compounds three separate nonlinear stages into the ceiling repeatedly	Back off Iron Drive/Flux/Hysteresis together, or raise Core Headroom, before reaching for Output.
Preset list looks different in the host's own menu vs. the plugin's in-GUI dropdown	These are two independent systems — an 81-entry host program list (27 recipes × 3 channel-mode banks) vs. an on-disk <code>.dih</code> folder with 27 entries plus anything saved later	Know which list you're browsing; the in-GUI dropdown reflects the disk folder, the host's own preset menu reflects program index.
Dual Mono channel mode doesn't sound any different from Stereo	Dual only changes routing when Dual Mono Link is also set to Independent	Set Dual Mono Link = Independent.
RESET doesn't return to "WP-R MKVI CANON" specifically	RESET loads index 0 of the alphabetically sorted on-disk preset list, not a pinned default	Select "WP-R MKVI CANON" from the dropdown directly.

9. Verified in this build

The parameter layout, DSP signal path, bus-layout logic, and preset system described above were read directly from `source/PluginProcessor.h`, `source/PluginProcessor.cpp`, `source/DIHFfield.h`, `source/PluginEditor.h`, and `source/PluginEditor.cpp` — not inferred or guessed. Version, plugin code, manufacturer code, and AU type came from `CMakeLists.txt`. No build, `auval` run, or DAW load was performed to write this manual.

The release folder shipped alongside this build

(`Releases/DMI_TRACK_WP-R_369_Signature_Channel_RELEASE_20260821/Validation_Logs/`) contains pre-existing signing/notarization logs, read (not generated) while writing this manual: the installer `.pkg` is signed with a Developer ID Installer certificate (William Slocumb, CQHHA4S27B) and was submitted to Apple's notary service, which returned status Accepted, after which the notarization ticket was stapled successfully. The same folder's `pkg_gatekeeper.log`, however, records a separate `spctl` assessment of rejected — Unnotarized Developer ID. These two logs disagree; re-run `spctl --assess -vv` on the `.pkg` yourself before redistributing this dated build rather than trusting either log in isolation.